

A Statistical Model of Auroral Ion Precipitation

2. Functional Representation of the Average Patterns

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In the study of Hardy et al. (1989) the average pattern of auroral ion precipitation was determined as a function of corrected geomagnetic latitude, magnetic local time (MLT), and Kp using data from the SSJ/4 detectors on the satellites of the Defense Meteorological Satellite Program (DMSP). One product of this study was a set of hemispheric maps of the integral number flux and integral energy flux of the precipitating ions for each of seven levels of activity as defined by Kp . In order to facilitate the use of these maps in magnetospheric, ionospheric, and thermospheric models, they have been fit to a functional form. To accomplish this, the latitudinal variation in the integral energy and number flux in each of 48 half-hour MLT bins of the original statistical study, for each Kp level, was fit to a general form of the Epstein function. This functional form was chosen since it fits well the asymmetric shape of the latitudinal profile of the integral energy and number flux. The Epstein function fits to the latitudinal profiles require six variables in each MLT zone. To further reduce the number of variables needed to specify a map, the MLT variation of each of the six variables was expanded as a sixth-order Fourier series with 13 coefficients. The resulting fit reproduces well the original statistical data. The distribution of differences between the original data and the fit normalized to the original data shows agreement that is generally better than 25% for both the integral energy and number flux.

1. INTRODUCTION

In recent years the Geophysics Laboratory has completed a series of studies of the average pattern of high-latitude electron and ion precipitation as a function of geomagnetic activity as measured by Kp and as a function of solar wind and interplanetary magnetic field (IMF) parameters [Hardy et al., 1985, 1987; Brautigam et al., 1988; Hardy et al., 1989]. In these studies we have determined the average electron and ion spectra in a series of spatial elements in corrected geomagnetic latitude and magnetic local time for different levels of geomagnetic activity and solar wind/IMF conditions. From the average spectra we calculate the integral energy flux, integral number flux, and average energy in each spatial element and in this way construct hemispheric maps for these quantities.

While these maps provide important information on the character of auroral precipitation, they do pose a problem to those who wish to use them for modeling. In most cases each map has 1440 spatial elements for each separation by activity or solar wind/IMF conditions. To include such relatively large arrays in other models can be cumbersome. In the past, to aid in the application of the results, we have fit the hemispheric maps of electron precipitation and their induced conductivities as a function of Kp to a simple functional form [Hardy et al., 1987]. This has been possible due to the smooth and highly regular variation in the integral quantities

of the electron spectrum as a function of both corrected geomagnetic latitude and magnetic local time.

In our previous work [Hardy et al., 1989] we presented results on the average pattern of auroral ion precipitation as a function of Kp . Two important results of this study were that globally the integral energy input from precipitating ions is a significant fraction of that for electrons and that for some magnetic local times and latitudes, namely on the evening side of the oval, the integral energy flux of ions can equal or exceed that for electrons. These results argue that the energy input from ions needs to be included in models of ionospheric and thermospheric dynamics.

With this in mind we have fit the results of the Hardy et al. [1989] paper in a manner similar to that used in the Hardy et al. [1987] paper. In this paper we report on the results of this fitting. In section 2 we briefly review the ion model. In section 3 the fitting procedure is described. In section 4 the results of the fitting and the degree to which the fitting reproduces the original data are discussed. Section 5 presents a guide to the reconstruction of the model from the coefficients.

2. THE PRECIPITATING ION STATISTICAL MODEL

The statistical data base that was the basis for the functional fits was created using data from the SSJ/4 particle spectrometers flown on the F6 and F7 satellites of the Defense Meteorological Satellite Program (DMSP). These spectrometers determined the electron and ion spectrum once per second in the local satellite zenith direction over 20

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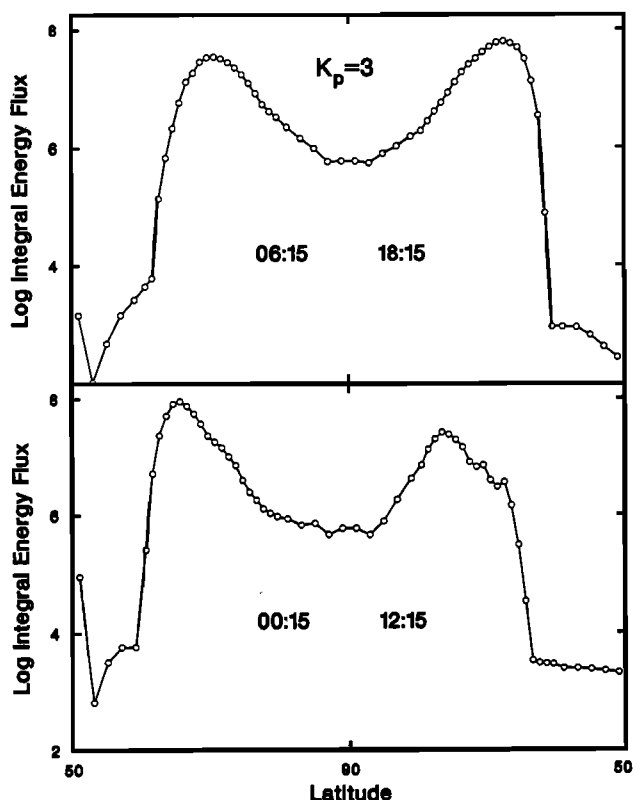


Fig. 1. Latitudinal cuts along the 0615–1815 and 0015–1215 magnetic local time meridians of the logarithm of the integral energy flux of the precipitating ions for $K_p = 3$ as determined in the statistical study of Hardy *et al.* [1989]. The latitude range is from 50° to 90° in corrected geomagnetic latitude, and the energy flux is plotted in units of $\text{keV} (\text{cm}^2 \text{ s sr})^{-1}$.

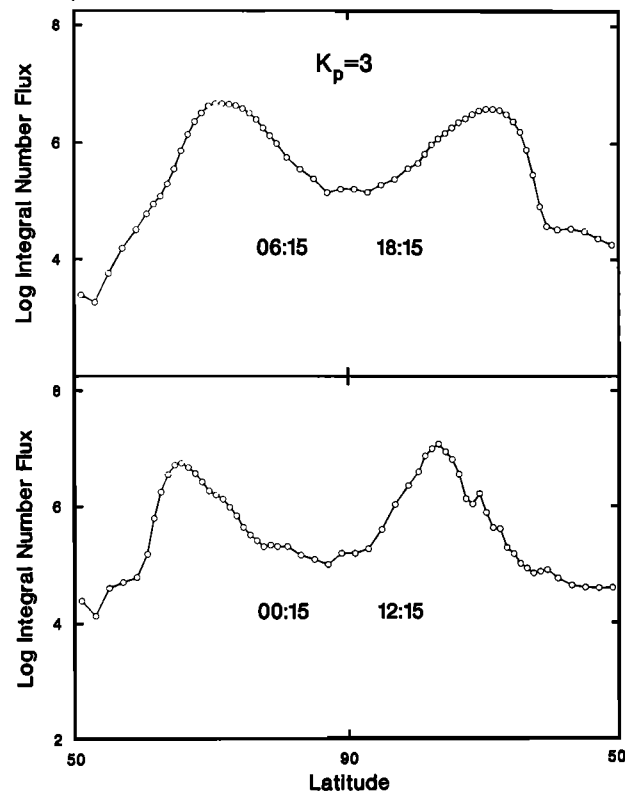


Fig. 2. The same as in Figure 1 for the integral number flux of precipitating ions. The number flux is plotted in units of $\text{ions} (\text{cm}^2 \text{ s sr})^{-1}$.

channels spanning the energy range from 30 eV to 30 keV. For the study, the high-latitude region was divided into a series of spatial bins in corrected geomagnetic latitude (CGL) and magnetic local time (MLT). There were 48 half-hour divisions in MLT and 30 divisions in CGL between 50° and 90° : 2° divisions between 50° and 60° CGL and 80° and 90° CGL and 1° between 60° and 80° CGL. One such spatial matrix was created for seven levels of activity as defined by K_p : one matrix for each whole value of K_p between 0 and 5 and one final matrix for $K_p \geq 6$. Using 1 year of data from both the F6 and F7 satellites, the average ion spectrum was determined in each spatial element for all levels of K_p . The large size of the DMSP data set ensured that a reasonable number of individual 1-s spectra occurred in each spatial element traversed by the satellite such that a statistically meaningful average spectrum could be determined in each such spatial element. Where appropriate, the average ion spectrum was extrapolated beyond the 30-keV upper energy limit of the detector to provide a better specification of the total precipitating ion number and energy flux. (For more details on the instrument, see Hardy *et al.* [1984]. For more details on the model, see Hardy *et al.* [1989].)

The integral number flux (JTOT), integral energy flux (JETOT), and average energy (EAVE) in each spatial element were calculated by integrating over the average spectrum using the equations

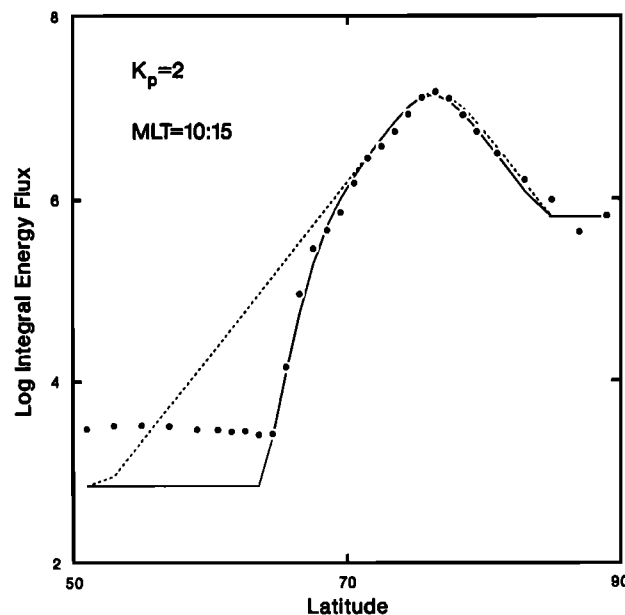


Fig. 3. A comparison of the fit to the data for the simple Epstein function and the more general form of the Epstein function. Solid circles are the data for the logarithm of the integral energy flux in units of $\text{keV} (\text{cm}^2 \text{ s sr})^{-1}$, for $K_p = 2$, for the MLT zone from 1000 to 1030 hours, and for corrected geomagnetic latitudes between 50° and 90° . The dashed line gives the best fit using the simple Epstein function, and the solid line gives the best fit for the more general Epstein function.

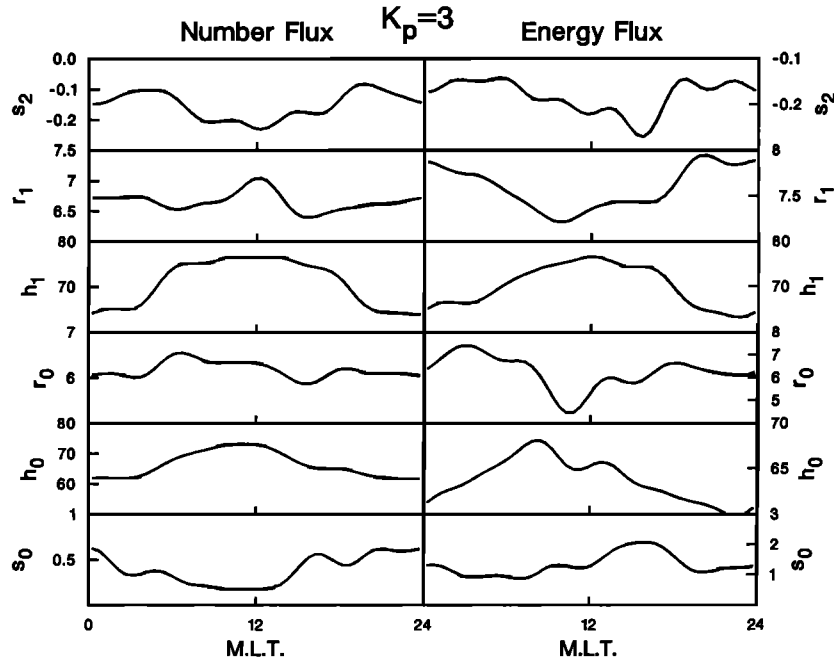


Fig. 4. The MLT variation in the six fit parameters used in the general Epstein function for the integral number flux (left-hand panels) and the integral energy flux (right-hand panels) and the magnetic activity, $K_p = 3$. The six parameters are S_0 and S_2 , the slopes of the Epstein function toward the equatorward and poleward edges of the oval, respectively; h_0 and h_1 , the latitudes at the first and second breakpoints; and r_0 and r_1 , the value of the Epstein function at the first and second breakpoints.

$$JTOT = \int j(E) dE \quad (1)$$

$$JETOT = \int E j(E) dE \quad (2)$$

$$EAVE = JETOT/JTOT \quad (3)$$

where $j(E)$ is the differential number flux in units of ions $(\text{cm}^2 \text{ s sr eV})^{-1}$. The integrals were calculated over the average spectrum by a piecewise approximation. The calculations give JTOT in units of ions $\text{cm}^2 \text{ s sr}^{-1}$, JETOT in units of $\text{keV cm}^2 \text{ s sr}^{-1}$, and EAVE in units of keV.

Typical meridional cuts of the integral energy and number flux taken from the model are shown respectively in Figures 1 and 2. These cuts are along the dawn-dusk and noon-midnight meridians for the $K_p = 3$ case. These are plotted to illustrate that the latitudinal variation is smooth and regular, as well as slowly varying in MLT. Both cuts show a single maximum at each MLT with a drop-off for values at latitudes above and below the peak. As we will discuss in the next section, there are characteristic changes in slope for either quantity with latitude both above and below the peak.

3. FITTING PROCEDURE

In a previous study [Hardy *et al.*, 1987] the latitudinal profiles of the base 10 logarithm of the electron integral number flux and integral energy flux were represented in each $\frac{1}{2}$ -hour-long MLT zone by an Epstein function [Booker, 1977] of the form

$$e(h) = r + S_1[h - h_0] + [S_2 - S_1] \ln \left\{ \left[1 - S_1/(S_2 e^{-(h-h_0)}) \right] / [1 - S_1/S_2] \right\} \quad (4)$$

where

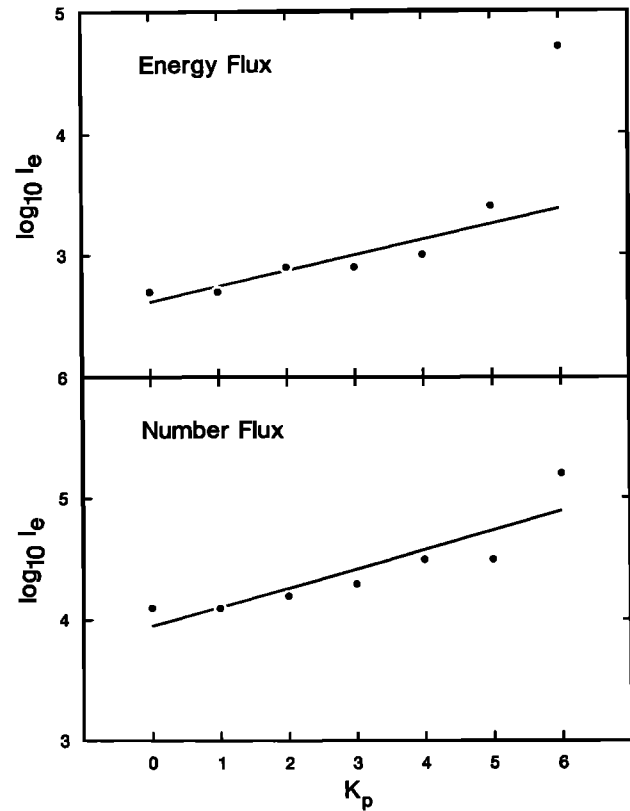


Fig. 5. The variation with K_p of the minimum level of the integral energy flux (top panel) and integral number flux (bottom panel) at latitudes below where the modified Epstein function fit applies. The solid circles give the data derived as explained in the text. The solid line is the least squares fit to the data.

TABLE 1. Fourier Coefficients for the Fit to the Magnetic Local Time Variation in Each of the Six Constants in the Epstein Function for Each Kp Range

	Integral Number Flux						Integral Energy Flux					
	r_0	h_0	h_1	s_0	r_1	s_2	r_0	h_0	h_1	s_0	r_1	s_2
$Kp = 0$												
c_0	6.0735	70.7046	74.3607	0.5626	6.5688	-0.1834	5.7511	67.4756	73.9120	1.6926	7.1132	-0.1758
c_1	0.0676	-4.7347	-5.4569	0.4117	-0.2088	0.0849	1.1791	-1.7068	-5.0821	-0.0322	0.0355	0.0123
c_2	0.0385	0.2563	-1.0669	0.1073	0.0649	-0.0586	-0.0659	-0.5087	-0.0316	0.1387	-0.0584	-0.0447
c_3	-0.1086	-0.6619	-1.0037	0.0723	-0.0137	-0.0072	0.1093	0.1568	0.1626	0.0694	-0.0314	-0.0132
c_4	0.0628	0.3734	0.5601	0.0195	0.0913	-0.0295	0.1002	0.4064	0.9433	-0.0633	-0.0218	-0.0212
c_5	0.0328	0.1372	0.0772	0.0133	-0.0705	0.0087	-0.1388	-0.4022	-0.3526	0.0547	-0.0026	0.0042
c_6	0.0024	0.2540	-0.0520	-0.0056	0.0527	-0.0136	-0.1005	-0.2477	0.0544	0.0523	0.0338	-0.0040
s_1	-0.0827	-0.4069	0.2668	0.0426	0.0111	0.0021	0.3902	0.7069	0.6570	-0.4330	-0.0594	0.0032
s_2	0.0807	0.4940	0.2659	0.0263	-0.0395	-0.0055	0.2499	0.1828	0.0486	-0.1219	0.0018	0.0178
s_3	-0.0765	-0.3480	-0.6087	-0.0529	-0.0093	-0.0009	-0.1009	0.2300	-0.5530	0.0436	0.0337	0.0045
s_4	0.0139	0.1891	-0.0287	-0.0852	0.0059	-0.0032	-0.1014	-0.5579	-0.3018	-0.0579	0.0241	0.0035
s_5	0.0504	0.0758	-0.2828	-0.0464	-0.0060	0.0033	-0.0703	0.1837	-0.0243	-0.0483	0.0250	0.0008
s_6	-0.0349	-0.2143	0.0979	0.0027	0.0080	-0.0105	-0.0257	0.0979	0.0164	0.0119	0.0321	-0.0019
$Kp = 1$												
c_0	5.9053	68.4378	72.9747	0.4932	6.5363	-0.1651	5.8306	65.8537	72.3306	1.7459	7.3263	-0.1818
c_1	-0.0968	-4.7863	-6.5478	0.3379	-0.1385	0.0561	0.9214	-1.7336	-5.5504	0.0086	0.2147	-0.0013
c_2	-0.1870	-0.2791	-1.5595	0.0922	0.0899	-0.0460	-0.0493	-0.9556	0.1021	-0.0252	0.0033	-0.0245
c_3	-0.0312	-0.2775	-0.4611	0.0800	-0.0495	-0.0140	-0.1558	0.2097	0.0737	0.2836	0.0165	-0.0118
c_4	0.1165	0.7060	1.1429	0.0259	0.0484	-0.0259	0.0838	0.1746	0.3974	-0.0047	-0.0218	0.0106
c_5	0.0121	0.1687	0.3651	0.0174	-0.0587	0.0163	-0.0549	0.2800	-0.1684	0.0082	-0.0252	0.0031
c_6	-0.0780	-0.1160	-0.0515	0.0172	0.0629	-0.0073	-0.0631	-0.0185	0.1947	-0.0288	0.0204	-0.0013
s_1	-0.0299	0.3443	-0.6087	-0.0303	0.0773	0.0000	0.8792	2.0586	0.7909	-0.9313	-0.0895	0.0088
s_2	-0.1102	-0.6511	-0.3428	-0.0164	-0.0316	-0.0049	-0.1005	-0.5218	0.2047	0.4387	-0.0035	-0.0050
s_3	-0.0316	0.0299	0.0311	-0.0026	-0.0175	-0.0040	0.0813	0.2187	-0.8707	-0.2428	0.0193	0.0113
s_4	0.0725	0.2290	0.5542	-0.0027	-0.0137	-0.0084	-0.0395	-0.2289	-0.3851	-0.0396	0.0183	0.0007
s_5	-0.0025	0.2316	-0.0589	0.0270	0.0082	0.0061	0.0328	0.1700	0.5390	-0.1407	-0.0290	-0.0034
s_6	-0.0375	-0.1297	0.1164	0.0374	0.0326	-0.0079	0.0428	0.2124	0.5290	0.0661	-0.0073	-0.0015
$Kp = 2$												
c_0	6.0361	67.2679	71.9932	0.4526	6.6232	-0.1559	6.3941	65.0188	72.0568	1.2547	7.4225	-0.1789
c_1	-0.0678	-5.1353	-6.3695	0.2446	-0.0767	0.0439	0.8898	-2.1418	-4.8412	0.1102	0.1676	0.0145
c_2	-0.1055	0.2245	-0.8987	0.0198	0.1221	-0.0345	-0.3394	-0.8018	0.6565	0.2185	-0.0946	-0.0182
c_3	-0.0897	-0.4851	-0.0985	0.0584	-0.0613	-0.0068	0.0462	0.3105	0.6312	0.1843	-0.0459	-0.0070
c_4	0.0778	0.4831	0.8770	0.0067	0.0321	-0.0158	-0.0170	0.0190	0.4389	0.0298	-0.0332	0.0014
c_5	0.0810	0.4403	0.1219	-0.0112	-0.0655	0.0141	-0.1832	-0.2340	-0.2156	0.0953	-0.0151	0.0047
c_6	-0.0205	0.0088	-0.5220	0.0111	0.0420	0.0000	0.0995	0.3475	0.2110	0.0596	-0.0123	0.0003
s_1	0.0982	1.0373	0.6313	-0.0777	0.0386	-0.0080	0.0458	1.5878	0.2083	-0.3183	-0.0820	0.0132
s_2	-0.1030	-0.9109	0.2883	0.0298	-0.0196	0.0037	-0.0485	-0.8014	-0.0295	0.1579	0.0052	-0.0052
s_3	-0.0595	-0.2820	-0.3917	0.0314	0.0272	0.0029	0.1351	0.2053	0.3723	-0.1712	-0.0638	-0.0020
s_4	0.0275	0.1571	0.3682	-0.0188	0.0092	-0.0047	0.0846	0.0865	0.5534	-0.0004	-0.0222	0.0031
s_5	0.0013	0.0356	0.4207	0.0191	-0.0136	-0.0046	-0.0295	0.0447	0.0464	-0.0858	-0.0143	-0.0030
s_6	-0.0259	-0.1248	0.0895	0.0286	0.0013	0.0126	-0.0983	-0.2953	-0.0738	-0.0039	0.0023	-0.0008
$Kp = 3$												
c_0	6.1780	66.3658	70.8355	0.3931	6.6621	-0.1518	6.2357	63.6903	70.1774	1.3190	7.5916	-0.1814
c_1	-0.0882	-5.4131	-6.8637	0.1736	-0.0313	0.0504	0.6770	-2.3486	-6.2159	-0.2011	0.3006	0.0364
c_2	-0.0426	0.6646	-0.8136	-0.0154	0.1492	-0.0265	-0.4664	-0.5513	0.1652	0.0418	-0.0036	-0.0100
c_3	-0.0209	-0.1186	0.8564	0.0490	-0.1051	-0.0184	-0.0469	0.2198	0.7147	0.1837	-0.0604	-0.0154
c_4	0.0901	0.5571	0.7227	0.0028	0.0476	-0.0043	0.0224	-0.1017	0.2763	-0.0245	0.0113	0.0079
c_5	-0.0289	-0.0202	-0.1785	0.0012	-0.0272	0.0093	-0.1516	-0.0532	-0.4385	0.0309	0.0215	0.0030
c_6	-0.0410	-0.1706	-0.4841	0.0227	0.0267	-0.0053	0.0271	0.1147	0.0514	-0.0421	0.0197	-0.0132
s_1	0.1380	1.5347	0.7320	-0.1220	0.0622	-0.0062	0.1812	2.1816	0.7639	-0.3473	-0.0859	0.0176
s_2	-0.1267	-1.2035	0.1830	0.0030	-0.0125	0.0011	0.4295	-0.0053	0.8098	0.2043	-0.0004	-0.0125
s_3	-0.0104	0.0752	-0.2095	-0.0207	0.0220	-0.0042	-0.0609	-0.0484	0.3498	0.0164	-0.0316	0.0029
s_4	0.0462	0.2787	0.0304	-0.0323	0.0127	0.0004	0.2810	0.5830	-0.0037	-0.0632	0.0445	0.0054
s_5	0.0079	0.0108	0.1931	0.0367	-0.0353	0.0050	-0.1924	-0.2247	0.2868	0.0118	0.0052	-0.0096
s_6	0.0016	0.0704	0.1581	0.0252	0.0119	-0.0061	0.0480	0.2563	0.3650	0.0589	-0.0093	0.0010
$Kp = 4$												
c_0	6.3494	65.1123	70.0659	0.3620	6.6308	-0.1510	6.3981	61.7691	68.5769	1.1444	7.6866	-0.1537
c_1	-0.2118	-5.9637	-7.8081	0.0512	0.1117	0.0632	0.2132	-4.1102	-5.8160	0.0098	0.4057	0.0158
c_2	0.0994	1.3884	0.3630	-0.0494	0.0932	-0.0180	-0.2670	-0.3000	0.8990	-0.1496	-0.0630	-0.0091
c_3	-0.0700	-0.3390	0.9560	0.0014	-0.1282	-0.0162	0.0959	0.5408	0.2507	-0.0439	0.0482	0.0014
c_4	0.1407	0.7818	0.4954	0.0031	0.0214	0.0043	0.0948	0.7203	0.0440	0.0286	0.0579	-0.0108
c_5	-0.0696	-0.0011	-0.9646	-0.0407	0.0196	0.0107	0.0139	0.2881	-0.5393	-0.2211	0.0047	0.0100
c_6	-0.0569	-0.2468	0.1221	0.0170	-0.0080	-0.0075	0.0373	0.0871	0.4974	0.0842	-0.0082	-0.0044
s_1	0.1358	1.6492	-0.2820	-0.1449	0.1368	0.0182	-0.3711	1.1527	1.0285	-0.0270	-0.1168	0.0104
s_2	-0.0699	-1.1400	0.5412	-0.0081	-0.0775	-0.0157	0.1387	-1.1598	0.1031	0.0545	0.0053	0.0029
s_3	-0.0071	-0.0055	-0.2018	-0.0108	0.0207	0.0009	-0.4767	-1.4005	-0.5264	-0.0133	0.0287	0.0014

TABLE 1. (continued)

	Integral Number Flux						Integral Energy Flux					
	r_0	h_0	h_1	s_0	r_1	s_2	r_0	h_0	h_1	s_0	r_1	s_2
$Kp = 4$ (continued)												
s_4	0.0191	0.0905	-0.0819	-0.0411	-0.0046	0.0182	0.1359	-0.0378	0.0298	-0.1336	0.0086	-0.0047
s_5	-0.0522	-0.2494	-0.0501	0.0174	-0.0062	-0.0127	-0.0397	-0.1296	-0.0862	0.0585	0.0067	0.0068
s_6	-0.0029	0.0425	0.3234	0.0054	0.0002	0.0092	-0.1453	-0.1340	0.3609	0.1615	0.0049	-0.0085
$Kp = 5$												
c_0	6.6795	65.2640	71.6299	0.2999	6.5068	-0.1744	6.6518	61.0858	67.8985	0.9525	7.7154	-0.1591
c_1	0.0648	-4.5485	-5.5767	0.0348	0.0845	0.0990	0.0721	-3.8113	-6.6252	-0.0354	0.4498	0.0414
c_2	0.1286	2.0447	0.5242	-0.0221	0.0650	-0.0354	-0.1840	0.9485	0.2066	0.1288	-0.0909	-0.0064
c_3	-0.2036	-1.2831	-0.8137	0.0086	-0.0421	0.0098	0.1949	0.4178	0.5874	-0.0501	-0.0944	0.0045
c_4	0.0569	0.1651	0.0542	-0.0153	0.0565	-0.0149	0.0344	-0.0165	0.4289	0.0374	0.0522	-0.0102
c_5	-0.0254	0.4365	0.3245	-0.0080	-0.0135	-0.0003	-0.1279	-0.2311	0.3884	0.0301	-0.0113	0.0116
c_6	-0.0191	-0.1781	-0.6756	0.0122	0.0524	0.0107	0.0164	-0.1698	0.2445	-0.0137	0.0458	-0.0107
s_1	0.1736	2.4377	1.9135	-0.0765	0.0159	-0.0187	-0.0920	2.7137	1.2924	-0.4044	-0.0398	-0.0056
s_2	-0.0362	-0.4969	0.2052	0.0041	-0.0625	0.0022	0.1932	-0.7834	-0.2050	0.0729	0.0440	0.0075
s_3	0.0325	0.2676	-1.4787	-0.0182	0.1157	0.0410	0.0452	-0.1720	-0.0715	-0.1239	0.0510	-0.0079
s_4	-0.0015	0.2652	0.7552	-0.0173	-0.0579	-0.0204	0.2531	0.8853	1.0292	0.0341	-0.0544	-0.0044
s_5	-0.0217	-0.1594	-0.4954	0.0247	0.0446	-0.0081	-0.0400	0.2123	-0.3239	-0.0703	0.0559	-0.0019
s_6	0.0029	0.0634	-0.8850	-0.0106	0.0532	0.0175	-0.1101	-0.2201	-0.4850	0.0357	0.0127	0.0019
$Kp = 6$												
c_0	6.7626	62.6294	72.6673	0.2483	6.4243	-0.1828	7.4840	61.4745	76.3270	0.7083	6.9225	-0.3192
c_1	0.1270	-3.1127	-0.9404	-0.0198	-0.1965	0.0678	0.9338	-1.0330	1.1911	-0.5087	-0.1453	0.0690
c_2	0.0498	2.0641	1.8835	0.0155	-0.0416	-0.0577	-0.3816	0.0337	0.2322	0.3716	0.0171	-0.0523
c_3	-0.0840	-1.0180	-1.1484	-0.0059	-0.0525	0.0283	-0.0695	-1.7417	-1.2274	-0.0224	0.1445	-0.0354
c_4	0.0571	0.6589	-0.3514	-0.0030	0.0967	-0.0124	0.1002	1.0161	0.1598	-0.0097	0.0225	-0.0848
c_5	0.0166	-0.4474	1.3156	-0.0208	-0.0997	0.0064	-0.0874	-0.0204	-0.3654	0.0285	0.0189	-0.0009
c_6	-0.0725	0.3518	1.1227	-0.0048	-0.0721	-0.0102	0.0058	-0.3218	-0.1122	0.0361	0.0077	-0.0026
s_1	0.0906	3.0632	1.4844	-0.0504	0.0337	0.0012	-0.1541	2.8077	1.4314	-0.2429	0.0110	0.1255
s_2	-0.0425	-1.4825	-0.9121	0.0126	-0.0452	0.0267	0.1115	-0.2483	2.9245	0.2706	-0.2730	-0.0694
s_3	0.0326	1.3137	0.5804	-0.0227	-0.0584	0.0041	-0.0186	0.3303	0.9907	-0.2187	-0.1641	-0.0289
s_4	-0.0600	-0.3794	0.0158	-0.0003	-0.0133	-0.0027	-0.0522	0.3713	0.0429	0.1065	-0.0204	-0.0324
s_5	0.0066	-0.6190	-0.6696	-0.0038	0.0830	0.0036	0.0235	-0.1971	-0.0725	-0.0120	0.0472	-0.0193
s_6	-0.0168	0.9286	-0.6205	0.0197	-0.0014	0.0240	-0.0627	0.4691	-0.3150	0.0252	0.0097	-0.0002

h corrected geomagnetic latitude in degrees;
 r maximum value of the quantity $\log_{10}$ integral energy or number flux;
 h_0 latitude of the maximum value at this MLT;
 S_1 slope of the curve for $h \ll h_0$;
 S_2 slope of the curve for $h \gg h_0$.

The value of this function is equal to r at $h = h_0$. At some distance above and below h_0 the functions become linear with slopes S_2 and S_1 , respectively. We found that this function fits the latitudinal asymmetric electron profiles better than the Gaussian fits used by Reiff [1984] and Simons *et al.* [1985]. Only the integral energy flux and integral number flux were fit since the average energy can be calculated from these two quantities.

Referring to Figures 1 and 2, one sees that the ion fluxes would also be rather poorly represented by a Gaussian. We have found, however, that the simple Epstein function of equation (4) is inadequate as well. The difficulty is illustrated by Figure 3. This figure shows the latitudinal profile of the integral energy flux for the 1000–1030 MLT zone and the $Kp = 2$ plotted as solid circles along with a dashed line giving a fit to the data by equation (4). As can be seen, the simple Epstein function reproduces the relatively symmetric peak quite well, but not the continuously increasing slope of the profile between the peak and the equatorward boundary.

This situation can be more adequately modeled by a generalization of the Epstein function that includes two breakpoints in the slope. The form of the function is

$$e(h) = r_0 + S_0[h - h_0] + [S_1 - S_0] \ln \{[1 + e^{h - h_0}]/2\} + [S_2 - S_1] \ln \{[1 + e^{h - h_1}]/[1 + e^{h_0 - h_1}]\} \quad (5)$$

where

r_0 the value of the quantity at breakpoint 1;
 h_0 the latitude of breakpoint 1;
 h_1 the latitude of breakpoint 2;
 S_0 the slope of the curve for $h \ll h_0$;
 S_1 the slope of the curve for $h_0 < h < h_1$;
 S_2 the slope of the curve for $h \gg h_1$.

Using this definition, the model would be specified as a function of latitude at each MLT by three slopes, two latitudes, and one value r_0 . We have found it more convenient, however, to choose a value for r_1 , the value of the quantity at breakpoint 2, and to calculate the intermediate slope S_1 by the requirement that $e(h_1) = r_1$. This gives

$$S_1 = [r_1 - r_0 - S_0(h_1 - h_0) + S_0 b_1 - S_2 b_2]/(b_1 - b_2) \quad (6)$$

with

$$b_1 = \ln \{[1 + e^{h_1 - h_0}]/2\} \quad (7)$$

$$b_2 = \ln \{2/[1 + e^{h_0 - h_1}]\} \quad (8)$$

The solid line in Figure 3 shows the fit to the modified Epstein function of equation (5).

ION ENERGY FLUX [top] AND NUMBER FLUX Kp=0

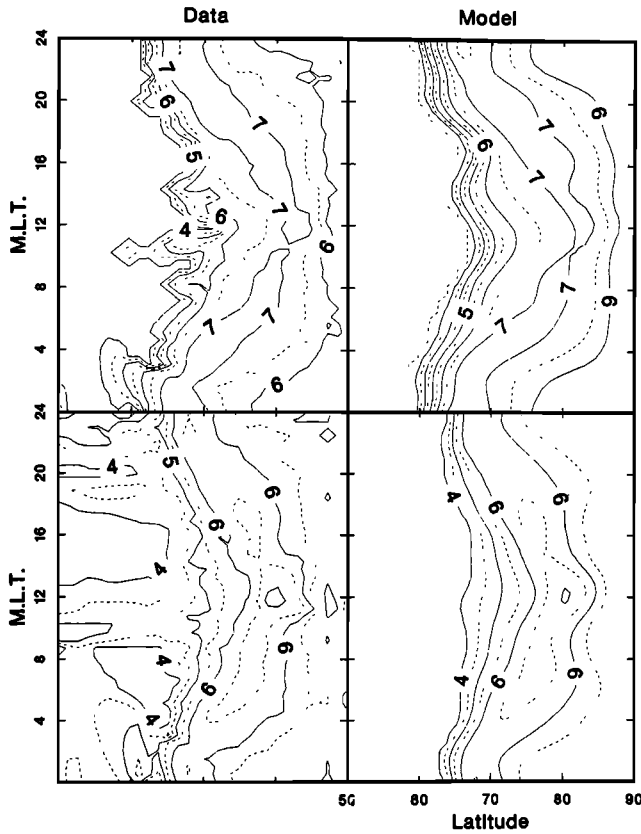


Fig. 6. Isocontour plots comparing the original data (left-hand panels) to the derived fits (right-hand panels) for $Kp = 0$. The two top panels show the comparison for the ion integral energy flux, and the two lower panels the comparison for the integral number flux. In each panel the isocontours are plotted versus MLT on the y axis and versus corrected geomagnetic latitude from 50° to 90° on the x axis. The isocontours are plotted at decade intervals.

Using the functional form of equation (5), the latitude and MLT dependence of the fluxes at each Kp division were determined as follows. First, for each MLT sector, two breakpoint latitudes h_0 and h_1 were selected, often referring to the preceding MLT sector results to ensure some degree of continuity in the coefficients between MLT values. The values of r_0 and r_1 were obtained from the data, and least squares fits were used to calculate S_0 and S_2 . The slopes were determined only for those points on either side of the maximum that appeared to be above the background level. This background level was also incorporated into the model, as will be discussed later. The breakpoint latitude points were adjusted at each MLT to obtain the best possible fit to the peak values of the fluxes. When this had been done for all 48 MLT divisions of a given Kp level, the 48 values of each of the six coefficients were fit to the function

$$a(T) = C_0^a + \sum_{n=1,5} C_n^a \cos [n\pi(T/12)] + S_n^a \sin [n\pi(T/12)] \quad (9)$$

where $a(T)$ is one of the parameters r_0 , r_1 , h_0 , h_1 , S_0 , or S_2 and C_n^a and S_n^a are the n th Fourier coefficients for that parameter. Here T is the magnetic local time in hours from 0

to 24. Next, using the coefficients regenerated from equation (9), the fit to the latitudinal profiles was examined again. In several cases, it was necessary to revise these regenerated coefficients and redo the Fourier fit in order to preserve the general quality of the fits at the individual MLT values in the model regenerated from the Fourier expansion. This iterative technique was necessitated in part by the fact that the latitudinal profiles were fit quite adequately in some regions of MLT by the single breakpoint function of equation (4), leaving some ambiguity as to the choice of the position of the second breakpoint. For consistency in these cases, we chose $r_0 < r_1$ and set h_1 near the maximum of the profile. In most cases it proved possible to preserve the quality of the fit in the Fourier expanded model, in that the results using coefficients calculated from the expansion were on average as good as those from the coefficients adjusted from the best fit of individual MLT profiles.

In Figure 4 the Fourier expansion in the six parameters in MLT are shown for the $Kp = 3$ case for both the integral number flux and integral energy flux. The MLT variation of the parameters for the $Kp = 3$ case is representative of the variation for the other Kp cases. For the integral number flux, S_0 and r_0 show a weak MLT variation. The largest values of S_0 , the slope toward the equatorward edge of the oval, are near local midnight MLT, and S_0 decreases to smaller values near noon. The value of the function at the first breakpoint, r_0 , has a maximum value at approximately 0700 MLT. The magnetic latitude of the first breakpoint, h_0 , varies smoothly with a maximum value near noon and a

ION ENERGY FLUX [top] AND NUMBER FLUX Kp=2

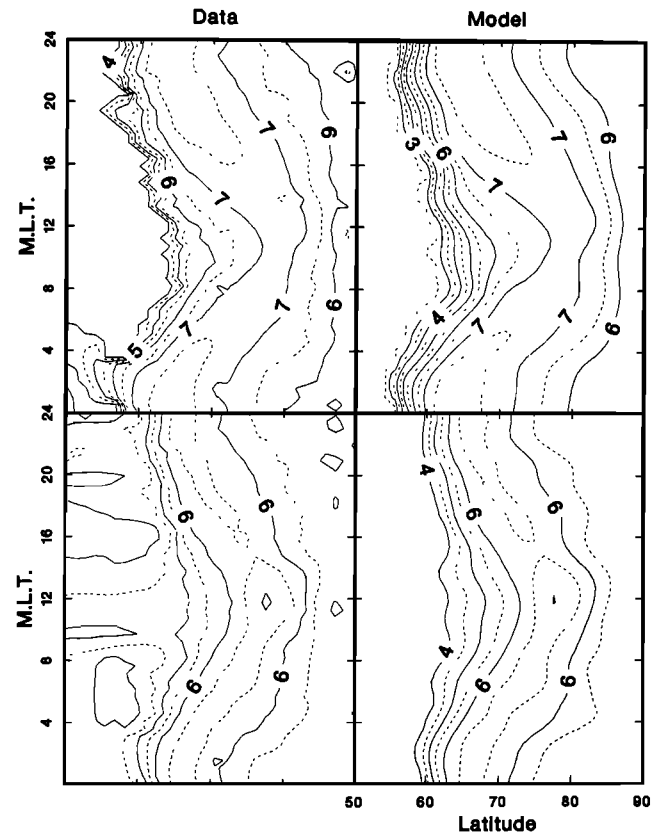


Fig. 7. In the same format as Figure 6 for $Kp = 2$.

minimum near midnight reflecting the well-known offset of the circular auroral precipitation pattern toward midnight. The functional value at the second breakpoint, r_1 , shows a maximum at noon due to the high level of integral number flux in the cusp. The magnetic latitude of the second breakpoint, h_1 , shows a strong MLT variation with a maximum at noon reflecting the change in latitude of peak integral number flux with MLT as one goes from midnight, where the peak results from central plasma sheet precipitation, to noon, where the peak results from precipitation in the cusp. The parameter S_2 , the slope toward the poleward edge of the auroral oval, has its smallest negative values at midnight and increases toward noon. The variation in the parameters for the integral energy flux shows similar general trends in MLT. The coefficients resulting from the fitting procedures are listed in Table 1. In the table we list the 13 coefficients for the Fourier expansion of each parameter for each level of Kp .

As stated above, the Epstein function approximation to the statistical model is only applicable above the approximately constant level to which the integral number and energy flux fall well equatorward and poleward of the central region of auroral ion precipitation. To determine this limiting level, the data well poleward and equatorward of the central precipitation region were examined separately. For each Kp separation for each of these regions, the residual root-mean-square (rms) deviation was calculated between the data of the statistical model and a specified constant level. This variation in this residual rms value was then determined as a function of the constant level chosen. For both integral

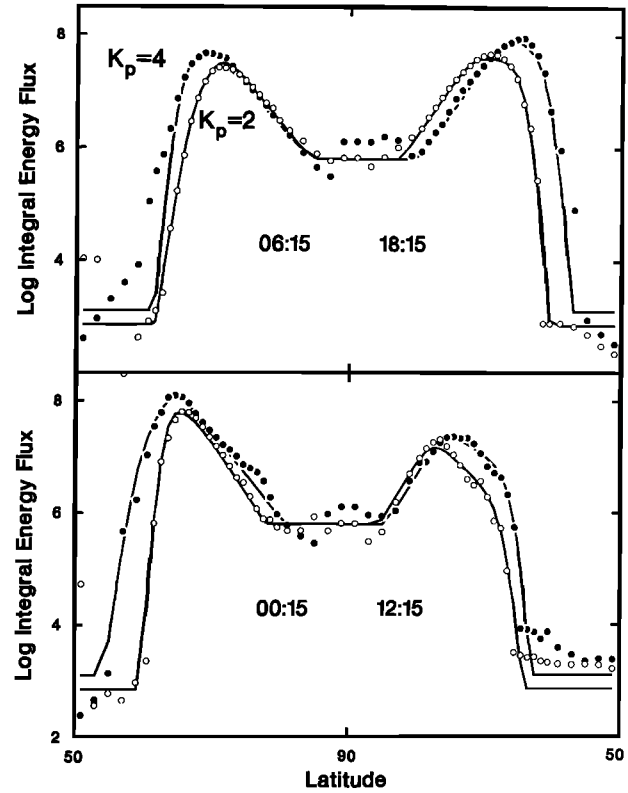


Fig. 9. A comparison of the fit to the original statistical data. The two panels show the comparison along the 0615–1815 (top panel) and 0015–1215 (bottom panel) MLT meridians for the integral energy flux for $Kp = 2$ and 4. Solid circles show the data for $Kp = 4$, and open circles the data for $Kp = 2$. The fits are shown by solid lines.

ION ENERGY FLUX [top] AND NUMBER FLUX $Kp=4$

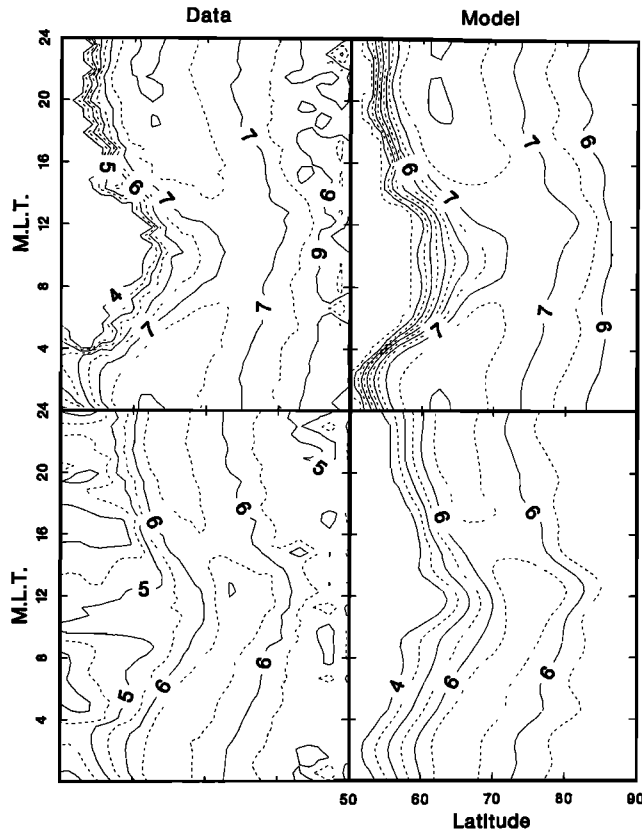


Fig. 8. In the same format as Figure 6 for $Kp = 4$.

number and energy flux and for all levels of Kp , the variation in the residual rms value with the constant level chosen displayed a single clear minimum. This minimum was chosen as the best limiting level for each Kp separation.

The poleward limiting values, l_p , determined in this way displayed no significant variation with Kp for either integral energy or number flux. The values were therefore chosen by averaging over Kp ; $1.41 \times 10^5 \text{ keV (cm}^2 \text{ s sr)}^{-1}$ and $6.31 \times 10^5 \text{ ions (cm}^2 \text{ s sr)}^{-1}$ for the integral energy flux and number flux, respectively. The limiting equatorward values, l_e , did show a trend, increasing with increasing Kp . This variation is shown along with a linear least squares fit to the points in Figure 5. Since the level plotted is that which minimizes the rms difference between that level and the data being fit, the level has no associated uncertainty. The fits give values for the limiting level of

$$l_e(\text{integral number flux}) = 8.32 \times 10^3 \times 10^{0.157Kp} \quad (10)$$

$$l_e(\text{integral energy flux}) = 4.17 \times 10^2 \times 10^{0.126Kp} \quad (11)$$

In the model, the Epstein function values should be used so long as they are greater than the levels specified above. At any latitude at which the values are less than these limits, the value should be set to these limits. In order to handle these limiting values, we evaluate equation (5) at h and assign the poleward limit l_p if $h > h_1$ and $e(h) < l_p$ or the equatorward limit l_e if $h < h_0$ and $e(h) < l_e$.

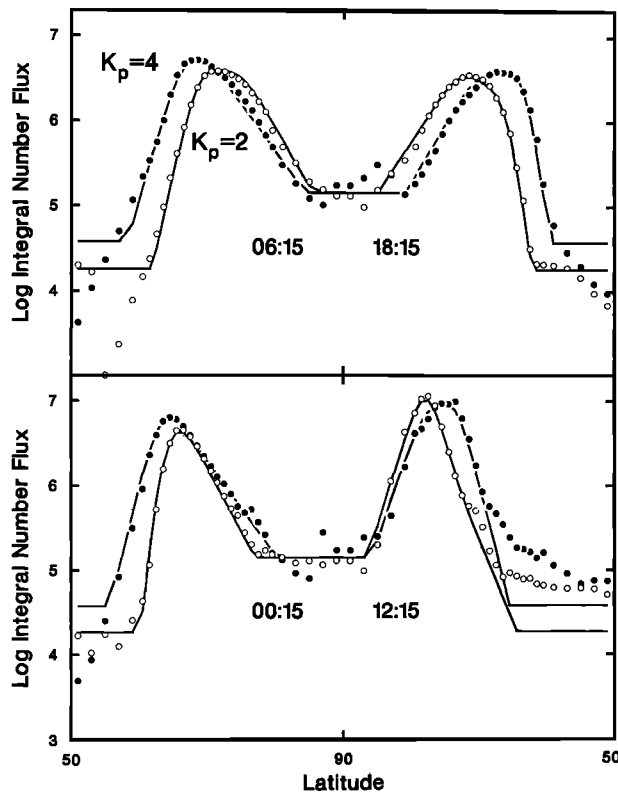


Fig. 10. In the same format as Figure 9 for the integral number flux.

4. QUALITY OF THE FUNCTIONAL FIT

In this section we examine the quality of the fit to the statistical data. In Figures 6–8 we show isocontours of the integral energy flux (top) and integral number flux (bottom), for the original raw data from the statistical model (left) and as reconstructed using the analytical expression derived (right). In the three figures the comparisons are shown for the $K_p = 0$, $K_p = 2$, and $K_p = 4$ cases in Figures 6, 7, and 8, respectively. One notes in all cases that the functional fit closely reproduces the original statistical model. As expected, the functional fit tends to smooth slightly the isocontours while faithfully reproducing in detail all of the major features of the statistical model.

This is shown in greater detail in Figures 9 and 10. In the figures, data have been plotted in the same format as in Figures 1 and 2 for $K_p = 2$ and $K_p = 4$ along with the fits for these data. One notes that despite significant variations in the latitudinal profile with local time, the fit provides a very close approximation to the original data. The quality of the fit is uniform over latitude except for a slight tendency for the fit to deviate from the data to a larger degree at latitudes well above and below the latitude of the peak integral energy or number flux. The original data do display some irregularities in latitude that are most likely the result of statistical fluctuations due to the small sample size in a number of latitude and MLT bins or due to a small level of residual noise in the input data. As such, the inability of the functional fit to follow these small-scale variations is not a failure of the fit but represents a reasonable smoothing out of these small-scale irregularities.

To quantify the quality of the fit, we calculated the

distribution of differences between the fit and the original data for each K_p map separately for the integral number flux and integral energy flux. Although the fit was performed to the logarithm of the integral energy and number flux, the differences were calculated from the actual values, i.e., $10^{e(h)}$. Each difference was divided by the model value to give the percentage differences. Differences were calculated only for values above the minimum values for which the fit applied. Representative examples of these distributions are shown in Figure 11, where the $K_p = 2$ case for both integral energy and number flux is plotted. For both integral energy flux and number flux the distributions are relatively narrow and symmetric about zeros, the full width at half maximum (FWHM) of the distribution being 0.24 ($\pm 12\%$) and 0.32 ($\pm 16\%$), respectively. The standard deviations for the distributions are larger, but these values are strongly affected by the low-frequency wings to the occurrence distribution. We interpret these wings as resulting from a few noise artifacts in the original data. As such, they do not represent a failure of the model to fit the data. Outside of the wings, the occurrence distribution is approximately Gaussian. For higher and lower activity the distribution broadens slightly with FWHMs at $K_p = 0$ and $K_p = 6$ of 0.40 and 0.44 (± 20 and $\pm 22\%$) and 0.48 and 0.52 (± 24 and $\pm 26\%$) for the integral number flux and integral energy flux, respectively. For $K_p = 2$ and 3, the most probable K_p , the number of data points in the statistical averages is largest. The number decreases for increasing and decreasing activity. We interpret the broadening of the “error” distribution as arising

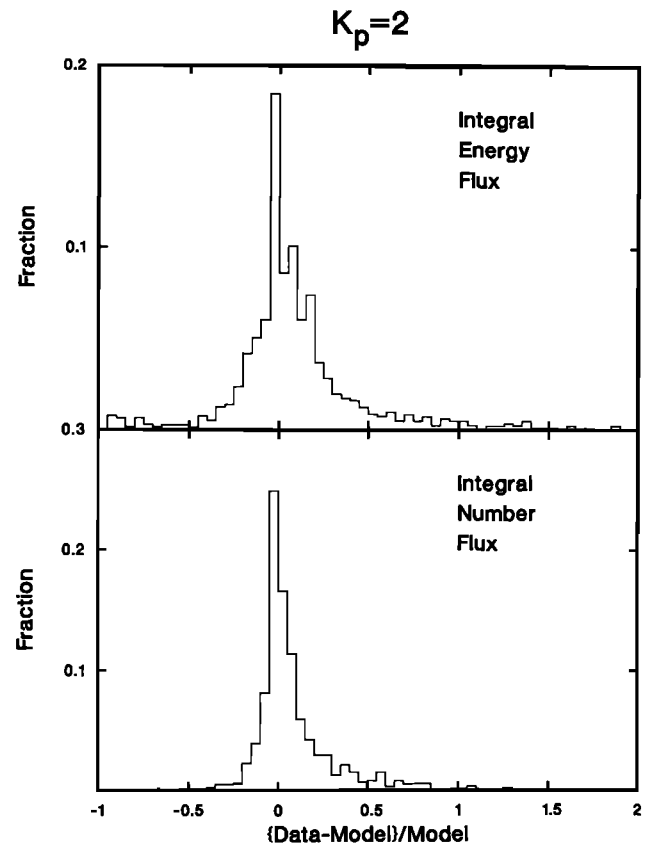


Fig. 11. The distribution of percentage differences between the original statistical data and the fit for $K_p = 2$ for the integral energy flux and integral number flux.

from increased fluctuations in the data due to decreasing sample size. For this type of "error" the fit acts to smooth out these residual fluctuations and therefore provides a better approximation to the real behavior than the original averages.

5. RECONSTRUCTION OF THE MODEL

The functional fit described in this paper provides a representation of the statistical model of auroral ion precipitation [Hardy *et al.*, 1989] and offers a simple means for reconstructing the hemispheric precipitation pattern without storing the large data matrices of the original model. A software subroutine that does the reconstruction is available from the Geophysics Laboratory, although it is a straightforward process to write such a subroutine from Table 1 of this paper and the definition of the Epstein function. In this process, equation (9) is used first to regenerate r_0 , r_1 , h_0 , h_1 , S_0 , and S_2 at the desired MLT value. Next, equations (6)–(8) are used to regenerate S_1 at the chosen MLT. As the next step, equation (5) is used to calculate the value of the function at any desired latitude. If values for intermediate Kp values are desired, this should only be done by interpolation of the calculated $e(h)$ values for different Kp levels. Interpolation of the coefficients themselves in Kp yields unpredictable results.

The reconstructed model can be used to specify for any corrected geomagnetic latitude, MLT, and Kp level the average integral energy flux, integral number flux, and average energy for the precipitating ions. It can also specify the approximate position of the equatorward boundary of the auroral oval as set by the ion precipitation. Since the level of energy flux of the ions is, at all activities, a significant fraction of the energy flux of the electrons and, for some latitudes and MLTs, equals or exceeds the electron energy flux, such a model can provide an important specification of the energy input to models of the ionosphere and thermosphere.

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REFERENCES

- Booker, H. G., Fitting of multi-region ionospheric profiles of electron density by a single analytic function of height, *J. Atmos. Terr. Phys.*, **39**, 619–626, 1977.
- Brautigam, D. H., M. S. Gussenhoven, and D. A. Hardy, The IMF B_2 and solar wind dependence for precipitating ion hemisphere energy flux, *Adv. Space Res.*, **8**, 9–12, 1988.
- Hardy, D. A., L. K. Schmitt, M. S. Gussenhoven, F. J. Marshall, H. C. Yeh, T. L. Schumaker, A. Huber, and J. Pantazis, Precipitating electron and ion detector (SSJ/4) for the block 5D/flights 6–10 DMSP satellites: Calibration and data presentation, *Rep. AFGL-TR-84-0314*, Air Force Geophys. Lab., Hanscom Air Force Base, Mass., 1984.
- Hardy, D. A., M. S. Gussenhoven, and E. Holeman, A statistical model of auroral electron precipitation, *J. Geophys. Res.*, **90**, 4229–4248, 1985.
- Hardy, D. A., M. S. Gussenhoven, R. Raistrick, and W. J. McNeil, Statistical and functional representations of the pattern of auroral energy flux, number flux, and conductivity, *J. Geophys. Res.*, **92**, 12,275–12,294, 1987.
- Hardy, D. A., M. S. Gussenhoven, and D. Brautigam, A statistical model of auroral ion precipitation, *J. Geophys. Res.*, **94**, 370–392, 1989.
- Reiff, P. H., Models of auroral zone conductances in *Magnetospheric Currents*, *Geophys. Monogr. Ser.*, vol. 28, edited by T. A. Potemra, pp. 180–191, AGU, Washington, D. C., 1984.
- Simons, S. L., P. H. Reiff, R. W. Spiro, D. A. Hardy, and H. W. Kroehl, A comparison of precipitating electron energy flux on March 22, 1979, with an empirical model: CDAW 6, *J. Geophys. Res.*, **90**, 2727–2734, 1985.
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